

DEAD WALLS

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UI-UX Decision System

CONTINUE

SETTINGS

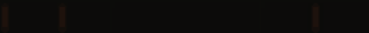
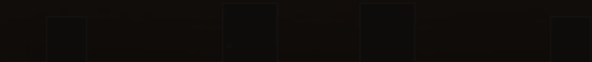
Research synthesis, locked product decisions, complete UI Toolkit architecture, surface-level interaction rules and live validation evidence.

FULL UI TOOLKIT

WORLD-FIRST

PC + LANDSCAPE MOBILE

ENGLISH-FIRST



The decision

What is now authoritative for Dead Walls

FULL UI TOOLKIT OWNERSHIP

01

Every player-facing menu, HUD, drawer, graph, modal and feedback layer is presented by UI Toolkit. The old Canvas remains only as a hidden behavior and serialized-data bridge during migration.

REWORK - NOT A PORT

02

The previous UI is not a visual, layout or styling reference. Information architecture and interaction design are rebuilt from the research model and the game's actual combat-economy workload.

WORLD-FIRST COMMAND UI

03

The battlefield stays readable. Persistent UI uses the edges, management uses side drawers, and only high-consequence systems take the full screen. Warm stone, bronze and restrained gold replace generic sci-fi chrome.

SIMULATION CONTINUITY

04

Economy, Barracks, Arrow Supply, Castle Heart and Technology do not pause or slow the simulation. Urgent wall, hostile and supply context remains visible inside full-screen management.

NON-NEGOTIABLE

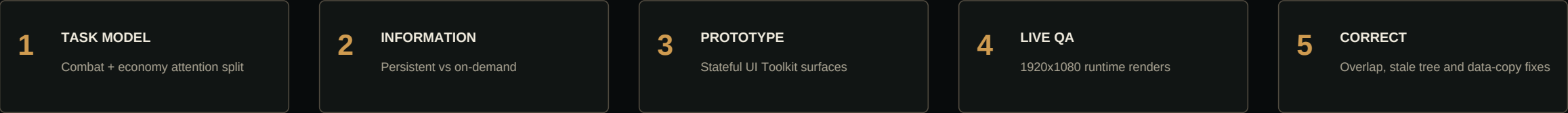
English-first. 1920x1080 reference. Automatic pointer / gamepad / touch presentation. Tech Tree remains a dynamic prerequisite graph. Main menu keeps only the approved day-dusk-night-dawn background loop.

A scientific evaluation frame

Evidence is translated into decisions, then validated in the live game

FRAME	QUESTION	DEAD WALLS APPLICATION
ISO 9241-11	Can players achieve goals effectively, efficiently and satisfactorily?	Measure whether combat and worker decisions can be completed without losing battlefield awareness.
Nielsen heuristics	Is status visible, language familiar, control recoverable and error prevention explicit?	Every management surface exposes current state, action, cost, result and blocker reason.
NASA-TLX	Where do mental, temporal and effort loads peak?	Night siege is the peak: persistent HUD is reduced to information that changes immediate action.
WCAG 2.2	Are targets, focus, contrast and non-color cues robust?	Touch hit areas expand, gamepad focus is explicit, and state labels accompany color.

VALIDATION LOOP



What comparable games taught us

Patterns retained for function, rejected when they conflict with Dead Walls

REFERENCE	USEFUL PATTERN	DEAD WALLS DECISION
They Are Billions	Fast population/resource scanning under swarm pressure.	Keep compact resource rates and population permanently visible.
Age of Darkness	Strong day/night threat framing and defensive urgency.	A centered siege clock communicates phase, countdown and pressure copy.
Cataclismo	Readable wall-building context and defense-first framing.	Wall integrity and arrow readiness share a single threat cluster.
Diplomacy is Not an Option	Large-scale combat with accessible macro controls.	Drawers preserve the world while exposing exact management actions.
Thronefall	Low-clutter battlefield presentation.	No ornamental HUD loops; restrained panels and edge anchoring.
Against the Storm	Explicit consequences, blockers and production tradeoffs.	Every decision surface states cost, result and unavailable reason.
The Last Spell	Tactical build planning and readable upgrade commitment.	Full-screen graphs use inspector-before-purchase and clear node states.

RETAINED

- Edge-anchored status clusters and strong phase signaling.
- Exact costs, consequences, prerequisites and disabled reasons.
- Fullscreen planning only for graph-scale systems.

REJECTED

- Dense permanent dashboards that compete with the battlefield.
- Generic grey sci-fi panels, neon decoration and AI-slop ornament.
- Pausing management screens or hiding urgent siege information.

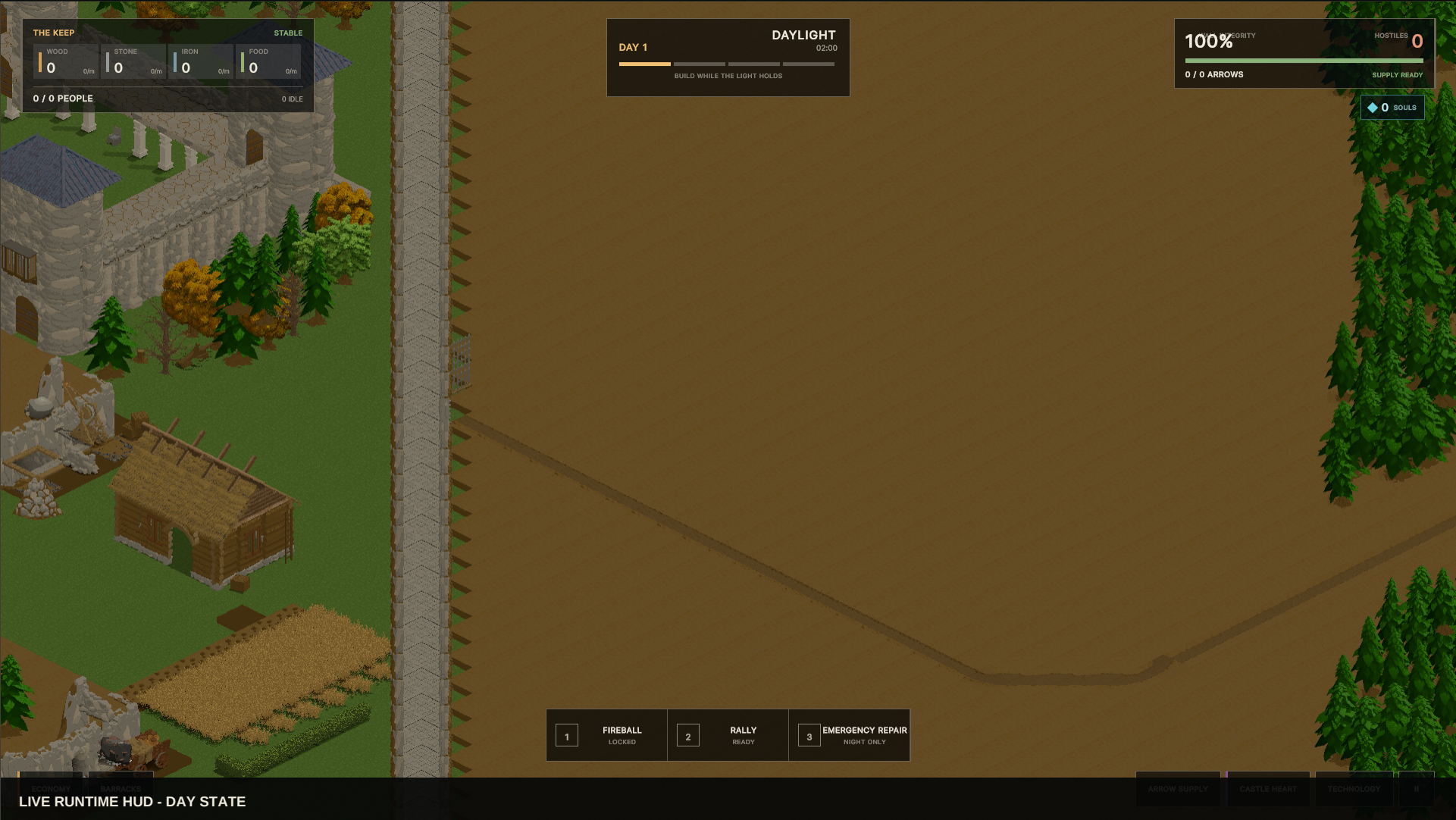
One player, two simultaneous fronts

The UI is organized around attention, not feature ownership



World-first command layer

Only information that changes immediate action remains persistent



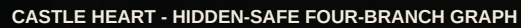
Manage without leaving the battle

Side drawers preserve spatial and temporal context



SURFACE	CURRENT STATE	ACTION	RESULT / BLOCKER
Economy	Available, workers, housing, target rate	+/- target, capacity, efficiency, housing	Updated labor allocation or exact resource/population blocker
Barracks	Archer roster, count, capacity	Recruit or retrain from live catalog	Formation change or cost/capacity reason
Arrow Supply	Current / max arrows and readiness	Crate, wagon, fill reserve, upgrades	Exact arrows received, wood cost or capacity blocker

Castle Heart and War Doctrine are dynamic graphs, not static menus



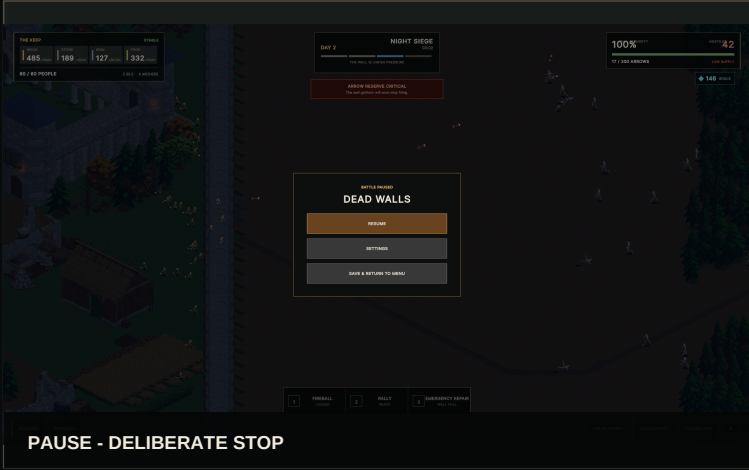
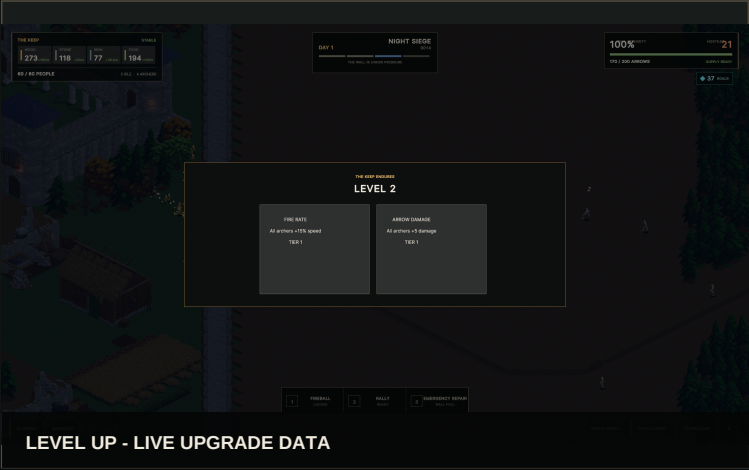
Consumes HeartGraphPresentation only. Unrevealed nodes never leak title, id or effect. Revealed choices show exact runtime-resolved values before purchase. Quantity controls support +1, +10 and MAX.



Depth is derived from prerequisites. Lanes, connectors and node states rebuild from the visible catalog. The graph recenters for small trees and supports pan, wheel/pinch-scale intent, zoom controls and inspector-first purchase.

High-consequence decisions get focus

The battlefield remains legible behind a consistent modal hierarchy



COUNCIL

Two mutually exclusive options, exact rich-text consequences, affordability and decision timer. The disabled state is driven by exact applicability, never approximate copy.

LEVEL UP

Cards are rebuilt directly from GameManager upgrade offers. The Toolkit no longer mirrors stale UGUI labels or placeholder object names.

PAUSE / SETTINGS / GAME OVER

One modal shade, one hierarchy and one visual language. Pause owns a pause lease; settings keep volume and tutorial reset; game over combines run result and meta shop.

Restrained medieval command language

Warm, legible and stateful - without decorative noise

WORLD BLACK

#080B0C

PANEL

#111514

PARCHMENT

#F0EBDE

BRONZE

#CF9B50

READY

#759F75

DANGER

#BE6650

DEFENSE

#7EA6C8

HEART MAGIC

#9A70B4

TOKEN	RULE	WHY
Typography	Condensed uppercase for labels; mixed-case sentence copy; strong numeric hierarchy.	Fast scanning without turning every line into a headline.
Spacing	8px rhythm; 44-52px pointer controls; expanded touch hit areas.	Consistency and landscape-mobile operability.
Motion	State transitions, cooldowns, damage, toast and soul flight only.	Movement always communicates cause or consequence.
Borders	One-pixel structure; two-pixel selected; four-pixel branch/state edge.	Hierarchy without heavy cards or glow.
Opacity	World remains visible under drawers; high-consequence modals darken more strongly.	Preserves context while controlling attention.

One document, three input modes

Presentation follows the last meaningful input automatically

POINTER + KEYBOARD

PC

Compact hit areas, visible 1/2/3 ability hints, hover feedback, wheel zoom and pointer drag for the doctrine graph.

GAMEPAD

PAD

The first meaningful action receives focus when a surface opens. Focus styling equals hover styling, and controller input changes the root presentation class.

TOUCH

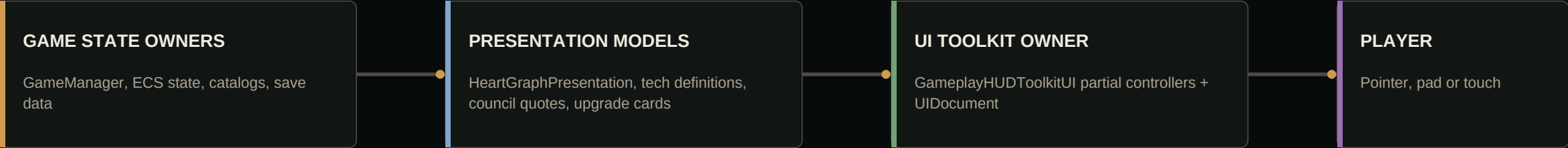
TOUCH

Landscape layout, expanded hit areas, compact responsive clusters and device-neutral onboarding copy. Management actions remain explicit rather than gesture-only.

ACCESSIBILITY CHECK	IMPLEMENTED CONTRACT
Status visibility	Day/phase/countdown, rates, wall, hostiles, arrows, souls and blockers are continuously or contextually visible.
Non-color cues	Every status color is paired with text, number, icon shape, disabled state or branch label.
Error prevention	Buttons disable from the real owner evaluation; exact blocker copy remains available in inspector or toast.
Recovery	Drawers close without commitment, graph view recenters, pause resumes, settings backtracks and tutorial reset requires confirmation.
Consistency	Shared surface headers, close/return actions, button states, inspectors, modals, toasts and scrollbar styling.

Presentation is new; behavior stays authoritative

The migration avoids rewriting gameplay logic inside UI code



PARTIAL	RESPONSIBILITY	DIRECT OWNER
Core	Lifecycle, visual-tree rebinding, HUD, input mode, surface coordination, legacy suppression	GameManager + runtime controllers
Management	Economy, Barracks, Arrow Supply rows and actions	Worker, Archer and Arrow runtime APIs
Graphs	Castle Heart graph, War Doctrine graph, inspectors, purchases, pan/zoom	Heart presentation + Tech catalog
Modals	Council, Level Up, Pause, Settings, Game Over, Meta Shop	Council quotes, upgrade cards, pause service, meta UI contracts
Feedback	Onboarding, toast mirroring, damage/tint and soul pickup flight	Existing events and exact world positions

What was verified

Live evidence, not static mockups

SURFACE / CONTRACT	LIVE CHECK	RESULT
Main menu	1920x1080 render; no labels for day/version/input; approved day-night background loop only	PASS
Persistent HUD	Resource + combat scan; clean world visibility; debug panel hidden by default	PASS
Drawers	Economy, Barracks and Arrow Supply rendered with live data and non-overlapping HUD clusters	PASS
Castle Heart	19-node hidden-safe graph; branch layout corrected after overlap QA	PASS
War Doctrine	Dynamic visible graph, connectors, centered small tree, pan/zoom and live inspector	PASS
Modals	Council exact options, direct-data level-up, pause/settings, game-over/meta shop and styled scroll	PASS
Runtime resilience	Visual tree rebinds after panel recreation/domain reload; old Canvas is hidden and non-interactive	PASS
Input	Root switched to input--touch; pointer/gamepad/touch classes and focus rules present	PASS

ACCEPTANCE CONTRACT

A surface is complete only when its real game-state data, actions, disabled reasons, input behavior, simulation policy and 1920x1080 visual hierarchy all agree. Screenshots in this document are runtime captures from NewGameScene and MainMenuScene, not design-only mockups.

Research sources

Primary standards, official guidance and evaluated game references

01	ISO 9241-11:2018 - Usability: Definitions and concepts	https://www.iso.org/standard/63500.html
02	Nielsen Norman Group - 10 Usability Heuristics	https://www.nngroup.com/articles/ten-usability-heuristics/
03	NASA - Task Load Index (NASA-TLX)	https://humansystems.arc.nasa.gov/groups/TLX/
04	W3C - Web Content Accessibility Guidelines 2.2	https://www.w3.org/TR/WCAG22/
05	Unity 6 Manual - Comparison of UI systems	https://docs.unity3d.com/6000.0/Documentation/Manual/UI-system-compare.html
06	They Are Billions	https://store.steampowered.com/app/644930/They_Are_Billions/
07	Age of Darkness: Final Stand	https://store.steampowered.com/app/1426450/Age_of_Darkness_Final_Stand/
08	Cataclismo	https://store.steampowered.com/app/1422440/Cataclismo/
09	Diplomacy is Not an Option	https://store.steampowered.com/app/1272320/Diplomacy_is_Not_an_Option/
10	Thronefall	https://store.steampowered.com/app/2239150/Thronefall/
11	Against the Storm	https://store.steampowered.com/app/1336490/Against_the_Storm/
12	The Last Spell	https://store.steampowered.com/app/1105670/The_Last_Spell/

FINAL PRODUCT POSITION

A readable medieval command interface built for simultaneous siege awareness and economic control.